Harmonising provenance information for biomedical data: the MIRAPIE approach applied to health research data in EHDS context

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Abstract

When exchanging data objects across institutions, means for comparison and interpretation of data objects from different sources are essentially required. Provenance information is a specific type of metadata describing the history of data objects across their life cycle. It allows assessment of data quality, reliability, or trustworthiness. In Europe, the European Health Data Space (EHDS) aims to foster health data reuse in research across national borders. The HealthDCAT-AP standard specifies metadata, including provenance information, to be provided along health data objects for the EHDS. The MInimal Requirements for Provenance Information Enrichment (MIRAPIE) project developed a harmonising framework for biomedical provenance information. This framework allows to set up provenance documentation from scratch but also to map already automatically collected provenance information. RO-Crate is an international project to support FAIR (Findable, Accessible, Interoperable, Reusable) metadata provision. In this workshop, we want to compare the national provenance approaches from Switzerland and Germany and apply the MIRAPIE framework to specify harmonised and EHDS-compliant provenance information and other metadata. RO-Crate is introduced as an option to sustainably publish such metadata.

Keywords

provenance, MIRAPIE, RO-Crate, European Health Data Space

1. Background

In Europe, different national solutions for health data sharing across institutions and for reuse in research are currently under development [1,2]. The EHDS, an infrastructure and legal framework for a secure exchange and common usage of health data, is designed to expand those national efforts to the whole European Union. This endeavor provides tremendous chances for secondary use of health data in research-based advancements in health care but also bears risks, ranging from simple misinterpretation of the data to severe data misuse. One way to mitigate these risks is the development and implementation of provenance information. Provenance generally refers to the documentation of the processing history of a data object along its life cycle. As a specific type of metadata, it is a key aspect of the FAIR principles (reusability indicator R1.2 “(meta)data are associated with detailed provenance”) [3] and provides a basis for assessing the quality, reliability, or trustworthiness of the described data object [4].

The HealthDCAT-AP standard is a health-focusing extension of DCAT-AP, developed and maintained by the European Union for the EHDS. DCAT-AP is an application profile of the W3C DCAT standard for describing datasets and distributions. HealthDCAT-AP was first developed as part of the EU project HealthData@EU Pilot and further developed in TEHDAS2. With Release 5, its

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management and further development have been taken over by the EU [5]. The HealthDCAT-AP vocabulary allows for a wide range of provenance annotations, from narrative descriptions to very generic association relationships. However, there is a lack of best practices for making these methods user-friendly.
The international community-driven MInimal Requirements for Automated Provenance Information Enrichment (MIRAPIE) project developed and implemented a framework to harmonise provenance representation by building on a purpose-driven minimal information approach [6,7]. This framework specifies provenance information according to seven W-questions summarized to the MIRAPIE question: “WHO did WHAT using WHICH methods, tools, and rules HOW, WHY, and WHEN with biomedical data obtained from WHERE?”. Following these questions, the MIRAPIE guideline provides actionable steps to implement provenance information for specific health datasets [8]. To cover biomedicine-specific requirements, a consensus data model and an accompanying upper-level provenance ontology have been designed [9]. Thus, provenance documentation setup from scratch and harmonisation of already existing provenance solutions are equally supported.
The RO-Crate project provides an integrated view through which you can see an entire research object; including methods, data, output [10]. It aims at supporting FAIR research by enabling to package and aggregate research artefacts with their metadata and relationships. This allows for facilitated persistent publication of metadata collections, thus leading to sustainably FAIR research projects.
While the EHDS provenance statement itself is not very specific [5], other concepts like purpose or role can also be addressed with MIRAPIE-compliant provenance documentation. In the current draft guideline on data minimisation, pseudonymisation, anonymisation and synthetic data, five of the seven W-keywords are specified as relevant [11]. We therefore hypothesise that harmonised FAIR provenance information supports fulfillment of the metadata requirements of the EHDS. Considering that the EHDS is going to further develop and most likely become more elaborated in the future, implementing the MIRAPIE approach for provenance documentation might sustainably prepare for the provision of interoperable provenance information documentation in EHDS context. Publishing the collected information using RO-Crate will increase FAIRness of data and metadata.

2. Workshop aim

This workshop aims to give an insight into different national solutions for provenance information documentation for health research data, into the provenance concept for the EHDS, and the harmonising MIRAPIE approach and the RO-Crate metadata collections. In the hands-on session, participants are invited to bring their own usage scenarios and we will apply the MIRAPIE guideline to their use case data.
Depending on the provenance maturity of the usage scenarios at session start, the MIRAPIE guideline can support to specify desired provenance information by answering the MIRAPIE question (maturity level 1), to choose a data model that suffices the specified level of detail and to apply the data model to the use case data (maturity level 2), and to map the resulting provenance information to the HealthDCAT-AP specifications of the EHDS (international interoperability).

Maximum Number of participants: 30
Anticipated duration: 4 hours, incl. break

3. Preliminary agenda

- Introduction round (15 min)
  - Motivation: why is provenance crucial for EHDS
- Impulse talks (60 min, moderation: tbd)
  - Challenges for FAIR provenance

        - MIRAPIE concept (Kerstin Gierend)
        - RO-Crate (tbd)
    - National initiatives
        - SPHN and its provenance concepts (Vasundra Touré)
        - Medical Informatics Initiative (Matthias Löbe)
    - European Health Data Space (EHDS)
        - HealthDCAT-AP core components and data provenance strategy (Matthias Löbe)
- Hands-on session part I (45 min): Best Practices
    - Application of MIRAPIE guideline to map example use cases and user projects to the EHDS (can be done by using the MIRAPIE data model or suitable alternatives; can be done in subgroups or all together)
- Break (15 min)
- Hands-on-session part II (45 min)
- Moderated panel / open discussion (60 min, moderation: tbd)

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